

Problem-Solution Fit

Date	29 June 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Section	Details
1. Customer Segment(s) (CS)	Banks, Financial Institutions, Bank Credit Officers, Loan Processing Officers, Financial Analysts
2. Problems / Pains + Frequency (PR)	Manual loan evaluation is slow, prone to human error, inconsistent, and becomes difficult during high application volumes. This occurs daily in banking operations.
3. Triggers to Act (TR)	Increase in loan applications, need for faster approvals, reducing loan defaults, improving customer satisfaction, and adopting AI-driven decision support.
4. Emotions Before / After (EM)	Before: Stressed, uncertain, overloaded. After: Confident, efficient, satisfied, and accurate in decision-making.
5. Available Solutions (AS)	Manual Evaluation: Accurate with expert review but slow and time-consuming. Rule-Based Systems: Fast but inflexible. Basic ML Models: Better than manual methods but may have lower prediction accuracy than XGBoost.
6. Customer Limitations (CL)	Limited time for evaluation, large workload, dependence on historical records, varying applicant profiles, and budget constraints for advanced software.
7. Behavior + Intensity (BE)	Officers review many loan applications every day and require quick, consistent decisions under time pressure.
8. Channels of Behavior (CH)	Online: Flask web application, IBM Cloud deployment, internal banking systems. Offline: Document verification, customer interviews, branch office operations.
9. Problem Root / Cause (RC)	Manual decision-making, increasing loan volumes, lack of intelligent automation, inconsistent assessment methods, and limited use of predictive analytics.
10. Your Solution (SL)	Develop Smart Lender using Machine Learning (XGBoost) integrated with a Flask web application to predict loan approval based on applicant details, enabling faster, accurate, and data-driven lending decisions.